

THE PROBLEM

Deviations go undetected until the FDA audit unravels everything

Clinical trials generate thousands of protocol deviations across hundreds of sites — missed visits, wrong doses, banned medications, late safety reports. Most sit unnoticed in disconnected systems and Excel logs until a regulator finds them.

\$50–100M

lost per rejected FDA submission

6–12 mo

typical drug-approval delay that follows

4–8 wks

between manual monitoring visits at a site

~5%

of a site's records a monitor can check per visit

Industry-wide, roughly 30% of new-drug submissions receive a Complete Response Letter — many citing data integrity or protocol-compliance issues that could have been caught earlier.

Deviation occurs



Logged in a spreadsheet



Unnoticed for 4–8 weeks



FDA audit finds it



Rejected submission

THE SOLUTION

TrialGuard: a hybrid rule engine + AI classifier

Deterministic rules do the detecting — auditable and traceable to the protocol. IBM watsonx.ai Granite adds judgment only where it can't undermine that: severity classification and CAPA drafting.

1

Detect

Rule engine checks every visit against the protocol: visit-window timing, dose accuracy, prohibited medications, required labs.



2

Classify

watsonx.ai Granite labels each deviation Major, Minor, or Administrative using few-shot prompts anchored to ICH E6(R2).



3

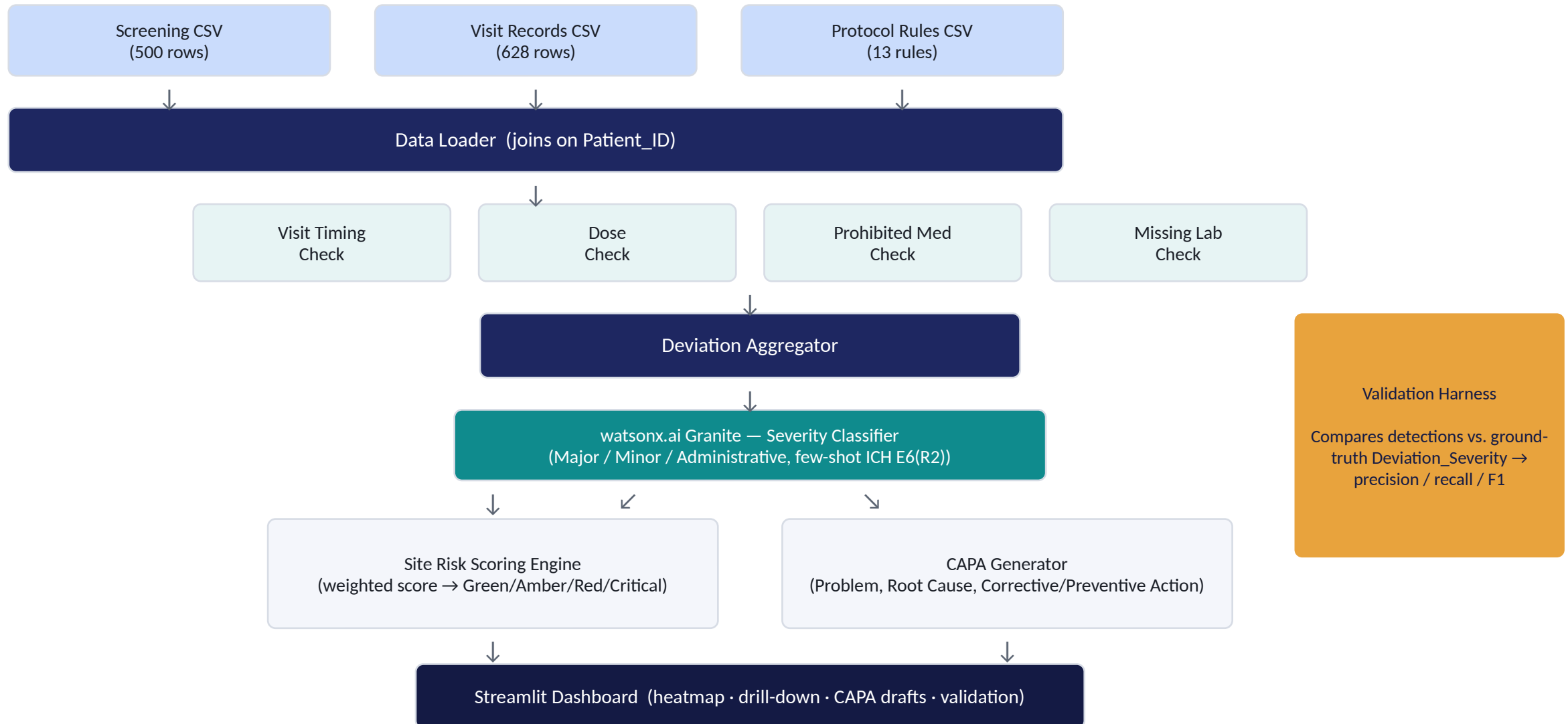
Score & Act

A weighted risk formula bands every site Green→Critical; watsonx.ai auto-drafts a structured CAPA for each Major/Minor finding.

Why hybrid, not pure LLM: regulators can't audit a pure-LLM system — outputs vary run to run and there's no formal spec to check against. Every TrialGuard deviation traces to a numbered rule; the AI only classifies and drafts, and a monitor can always override it.

ARCHITECTURE

A two-stage pipeline: rules detect, AI classifies and drafts



RESULTS

Validated against ground truth, not just demoed

Every visit record ships with a ground-truth Deviation_Severity label. The Validation Harness aligns rule-engine output against it and reports real precision and recall — evidence, not a claim.



25 of 26 labeled deviations detected — one missed case, fully traceable in the validation log

Dataset: 628 visit records · 26 labeled deviations · 20 Major, 5 Minor, 1 Administrative



Major (20)

Minor (5)

Administrative (1)

IBM Bob, used across the full development lifecycle

- **Plan mode** — Designed the system architecture and rule-engine structure
- **Code mode** — Wrote the deviation rules, watsonx.ai integration, risk scoring, CAPA generator, and Streamlit dashboard
- **Ask mode** — Generated AGENTS.md repo context and answered ICH E6(R2) domain questions
- **/review** — Final code review pass before submission
- **Custom “GCP-Auditor” mode** — Reviews outputs against ICH E6 principles

bob_sessions/

└─ palak/

exported Bob task-history markdown

└─ darshika/

exported Bob task-history markdown

└─ jahnvi/

exported Bob task-history markdown

└─ riya/

exported Bob task-history markdown

All Bob task-session exports are stored here, one folder per team member, and submitted for judging.